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Dept : Data Science

Course : MultiVariate Calculus

ASSIGNMENT # 04

LECTURE # 02

Line Integrals : When a curve $r(t) = g(t)\underline{i} + h(t)\underline{j} + k(t)\underline{k}$, $a \leq t \leq b$ passes through the domain of a function $f(x, y, z)$ in space the values of f along the curve are given by the composite function $f(g(t), h(t), k(t))$. If we integrate this w.r.t arc length from $t=a$ to $t=b$, we calculate the so called line integral of f along the curve.

- If eq. of the curve c in the xy -plane is given as $y = g(x)$, $a \leq x \leq b$ then the line integral.

$$\int_c f(x, y) ds = \int_a^b f[x, g(x)] \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- If c is given as $x = h(y)$, $c \leq y \leq d$ then.

$$\int_c f(x, y) ds = \int_c^d f[h(y), y] \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

- If c is given in parametric form $x = g(t)$, $y = h(t)$, $t_1 \leq t \leq t_2$ then

$$\int_c f(x, y) ds = \int_{t_1}^{t_2} f[g(t), h(t)] \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Ex) Evaluate $\int_C xy^3 ds$: where

i) C is the segment of the straight line $y = 2x$ from $(-1, -2)$ to $(1, 2)$.

ii) C is the circle having parametric equations
 $x = \cos t$, $y = \sin t$ $0 \leq t \leq \pi/2$.

Sol

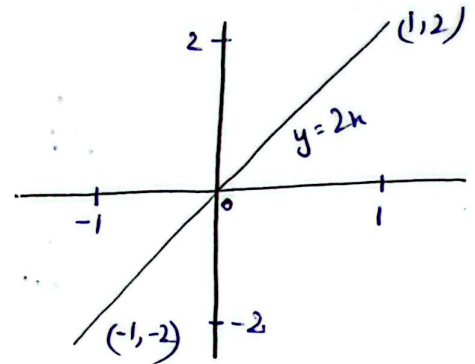
$$\begin{aligned}\int_C xy^3 ds &= \int_{-1}^1 x \cdot (2x)^3 \cdot \sqrt{5} dx \\&= 8\sqrt{5} \int_{-1}^1 x^4 dx \\&= 8\sqrt{5} \left. \frac{x^5}{5} \right|_{-1}^1 \\&= 8\sqrt{5} \left(\frac{1}{5} - \left(-\frac{1}{5} \right) \right) \\&= 8\sqrt{5} \left(\frac{2}{5} \right) = \frac{16}{\sqrt{5}}\end{aligned}$$

$$\frac{dy}{dt} = \cos t \quad \frac{dx}{dt} = -\sin t$$

$$\sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2}$$

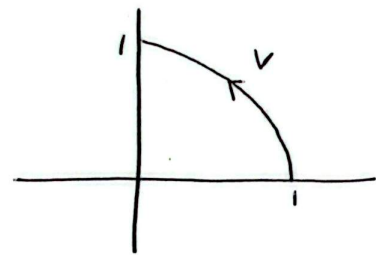
$$= \sqrt{\sin^2 t + \cos^2 t} = 1$$

$$\begin{aligned}\int_C xy^3 ds &= \int_0^{\pi/2} \cos t \sin^3 t \cdot 1 \cdot dt \\&= \int_0^{\pi/2} \sin^3 t \cdot \cos t \cdot dt \\&= \left. \frac{\sin^4 t}{4} \right|_0^{\pi/2} \\&= \frac{1}{4} - 0 = \frac{1}{4}\end{aligned}$$



$$y = 2x \Rightarrow \frac{dy}{dx} = 2$$

$$\sqrt{1 + \left(\frac{dy}{dx} \right)^2} = \sqrt{1 + 2^2} = \sqrt{5}$$



$$x = \cos t, \quad y = \sin t$$

$$x^2 + y^2 = r^2$$

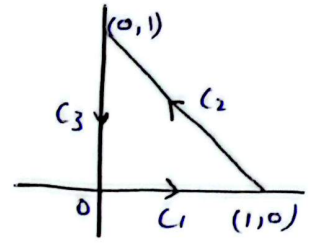
$$\cos^2 t + \sin^2 t = r^2$$

$$r^2 = 1$$

$$r = 1$$

Evaluate $\oint_C (x+y) ds$ where C is the contour of the triangle with vertices $(0,0)$, $(0,1)$ and $(1,0)$.

Sol: In this case the contour C consists of the line segments C_1 , C_2 and C_3 . We know that the equations of C_1 is $y=0$, C_2 is $y=1-x$ and C_3 is $x=0$.



$$\text{Thus, } \oint_C (x+y) ds = \int_{C_1} (x+y) ds + \int_{C_2} (x+y) ds + \int_{C_3} (x+y) ds \quad \text{--- (1)}$$

For C_1 , $y=0$ therefore $\frac{dy}{dx} = 0$, $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 1$.

$$\int_{C_1} (x+y) ds = \int_{x=0}^1 (x+0) \cdot 1 dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}$$

For C_2 , $y=1-x$ therefore $\frac{dy}{dx} = -1$, $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{2}$.

$$\begin{aligned} \int_{C_2} (x+y) ds &= \int_1^0 (x+1-x) \times \sqrt{2} dx \\ &= \sqrt{2} \left. x \right|_1^0 = \sqrt{2} (0-1) = -\sqrt{2}. \end{aligned}$$

For C_3 , $x=0 \Rightarrow \frac{dx}{dy} = 0$, $\sqrt{1 + \left(\frac{dx}{dy}\right)^2} = 1$.

$$\int_{C_3} (x+y) ds = \int_{y=1}^0 (0+y) \times 1 dy = \left. \frac{y^2}{2} \right|_1^0 = -\frac{1}{2}.$$

Put in eq. (1)

$$\oint_C (x+y) ds = \frac{1}{2} - \sqrt{2} - \frac{1}{2} = -\sqrt{2}.$$

Line integral along a space curve

$$x = g(t), \quad y = h(t), \quad z = k(t), \quad a \leq t \leq b$$

$$\int_C f(x, y, z) ds = \int_a^b f[g(t), h(t), k(t)] \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Evaluate $\int_C (xy + z^2) ds$ where C is the curve defined by $x = \cos t$, $y = \sin t$, $z = t$, $0 \leq t \leq \pi$

$$\begin{aligned} \text{Sol } \int_C (xy + z^2) ds &= \int_0^\pi (\cos t \sin t + t^2) \sqrt{2} dt & \frac{dx}{dt} &= -\sin t \\ &= \sqrt{2} \left[\frac{\sin^2 t}{2} + \frac{t^3}{3} \right]_0^\pi & \frac{dy}{dt} &= \cos t, \frac{dz}{dt} = 1 \\ &= \sqrt{2} \left(\left(0 + \frac{\pi^3}{3}\right) - (0 + 0) \right) = \frac{\sqrt{2} \pi^3}{3} & \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} &= \sqrt{2} \end{aligned}$$

Find the line integral of $f(x, y, z) = 2xy + \sqrt{z}$ over the helix $\underline{r}(t) = \cos t \underline{i} + \sin t \underline{j} + t \underline{k}$, $0 \leq t \leq \pi$

$$\int_C (2xy + \sqrt{z}) ds = ?$$

$$x = \cos t, \quad y = \sin t, \quad z = t.$$

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = 1.$$

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2}$$

$$= \sqrt{1+1} = \sqrt{2}.$$

$$\int_C (2xy + \sqrt{z}) ds = \int_0^\pi (2 \sin t \cos t + t^{1/2}) \sqrt{2} dt$$

$$= \sqrt{2} \left[\sin^2 t + \frac{t^{3/2}}{3/2} \right]_0^\pi$$

$$= \sqrt{2} \left[\left(0 + \frac{2}{3} \pi^{3/2}\right) - (0 + 0) \right]$$

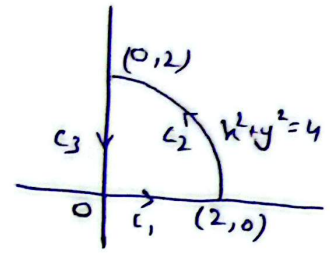
$$= \frac{2\sqrt{2} \pi^{3/2}}{3}$$

Evaluate $\oint_C (x^2 + y^2) ds$ where C is the contour of the triangle with vertices $(0,0)$, $(0,2)$, $(2,0)$

Sol we know that eqs. of C_1 is $y=0$,

C_2 is $x^2 + y^2 = 4$ and C_3 is $x=0$.

$$\oint_C (x^2 + y^2) ds = \int_{C_1} (x^2 + y^2) ds + \int_{C_2} (x^2 + y^2) ds + \int_{C_3} (x^2 + y^2) ds \quad \text{--- (1)}$$



For C_1 , $y=0 \Rightarrow \frac{dy}{dx} = 0$, $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 1$.

$$\int_{C_1} (x^2 + y^2) ds = \int_0^2 (x^2 + 0) dx \cdot 1 = \left. \frac{x^3}{3} \right|_0^2 = \frac{8}{3}.$$

For C_2 , $y^2 + x^2 = 4$, we use polar parameterization

$x = 2\cos t$, $y = 2\sin t$, where $0 \leq t \leq \pi/2$.

$$\frac{dx}{dt} = -2\sin t, \quad \frac{dy}{dt} = 2\cos t \Rightarrow \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 2.$$

$$\int_{C_2} (x^2 + y^2) ds = \int_0^{\pi/2} 4 \cdot 2 dt = [8t]_0^{\pi/2} = 8\left(\frac{\pi}{2}\right) = 4\pi.$$

For C_3 , $x=0 \Rightarrow \frac{dx}{dy} = 0$, $\sqrt{1 + \left(\frac{dx}{dy}\right)^2} = 1$.

$$\int_{C_3} (x^2 + y^2) ds = \int_2^0 y^2 \cdot 1 dy = \left. \frac{y^3}{3} \right|_2^0 = -\frac{8}{3}.$$

Put in eq. (1)

$$\oint_C (x^2 + y^2) ds = \frac{8}{3} + 4\pi - \frac{8}{3}$$

$$= 4\pi$$

LECTURE # 03

Second Type: If the equation of the curve C in the xy -plane is given as $y = f(x)$, $a \leq x \leq b$ then

$$\int_C P(x,y) dx + Q(x,y) dy = \int_a^b P[x, f(x)] dx + Q[x, f(x)] f'(x) dx$$

If C is given $x = g(y)$, $c \leq y \leq d$ then

$$\int_C P(x,y) dx + Q(x,y) dy = \int_c^d P[g(y), y] g'(y) dy + Q[g(y), y] dy$$

If C is in parametric form $x = f(t)$, $y = g(t)$, $t_1 \leq t \leq t_2$

$$\int_C P(x,y) dx + Q(x,y) dy = \int_{t_1}^{t_2} P[f(t), g(t)] f'(t) + Q[f(t), g(t)] g'(t) dt$$

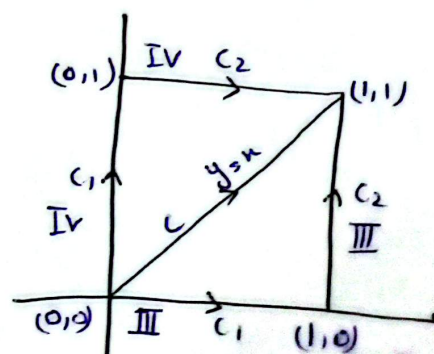
Evaluate $\int_C (1+x^2y) dx + 2xy dy$ from $(0,0)$ to $(1,1)$.

along the following paths C ,

- i) straight line from $(0,0)$ to $(1,1)$
- ii) Parabola $y = x^2$ from $(0,0)$ to $(1,1)$
- iii) line segment C_1 from $(0,0)$ to $(1,0)$ and then the line segment from $(1,0)$ to $(1,1)$
- iv) line segment C_1 from $(0,0)$ to $(0,1)$ and then the line segment C_2 from $(0,1)$ to $(1,1)$

Sol (i) Along the st. line $y=x$ we have $dy=dx$ while x varies from 0 to 1.

$$\begin{aligned} \int_C (1+x^2y) dx + 2xy dy &= \int_0^1 (1+x^2 \cdot x) dx + 2x \cdot x dx \\ &= \int_0^1 (1+x^3+2x^2) dx = x + \frac{x^4}{4} + \frac{2x^3}{3} \Big|_0^1 \\ &= 1 + \frac{1}{4} + \frac{2}{3} = \frac{23}{12} \end{aligned}$$



Eq of line passing through two points $(0,0)$ & $(1,1)$ is $y=x$

ii) Along the parabola $y = x^2$, we have $dy = 2x dx$, x from 0 to 1. (1)

$$\begin{aligned}\int_C (1+x^2y) dx + 2xy dy &= \int_0^1 (1+x^2 \cdot x^2) dx + 2x \cdot x^2 \cdot 2x dx \\ &= \int_0^1 (1+x^4 + 4x^4) dx = \int_0^1 (1+5x^4) dx \\ &= \left[x + x^5 \right]_0^1 = 1+1 = 2.\end{aligned}$$

iii) In this case

$$\int_C (1+x^2y) dx + 2xy dy = \int_{c_1} (1+x^2y) dx + 2xy dy + \int_{c_2} (1+x^2y) dx + 2xy dy \quad \text{--- (1)}$$

Along the line segment from $(0,0)$ to $(1,0)$

$y=0$, $dy=0$ while x from 0 to 1.

$$\int_{c_1} (1+x^2y) dx + 2xy dy = \int_{x=0}^1 dx = 1.$$

Along the line segment from $(1,0)$ to $(1,1)$

$x=1$, $dx=0$ while y varies from 0 to 1.

$$\int_{c_2} (1+x^2y) dx + 2xy dy = \int_{y=0}^1 2y dy = y^2 \Big|_0^1 = 1$$

$$\int_C (1+x^2y) dx + 2xy dy = 1+1 = 2.$$

iv) Along segment c_1 from $(0,0)$ to $(0,1)$

$x=0$, $dx=0$ while y from 0 to 1.

$$\int_{c_1} (1+x^2y) dx + 2xy dy = \int_0^1 0 dx = 0.$$

Along segment c_2 from $(0,1)$ to $(1,1)$

$y=1$, $dy=0$, x from 0 to 1.

$$\int_{c_2} (1+x^2y) dx + 2xy dy = \int_0^1 (1+x^2) dx = \left[x + \frac{x^3}{3} \right]_0^1 = 1 + \frac{1}{3} = \frac{4}{3}.$$

$$\int_C (1+x^2y) dx + 2xy dy = 0 + \frac{4}{3} = \frac{4}{3}$$

Green's Theorem

Let $P(x,y)$, $Q(x,y)$, $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ be continuous function

In a simply connected region R bounded by a simple closed curve C . Then

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where C is described in positive direction.

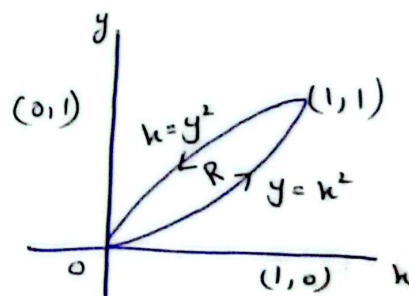
A simple closed curve is a closed curve which does not intersect itself anywhere.



Verify Green's theorem in the plane for

$\oint_C (2xy - y^2) dx + (x + y^2) dy$ where C is a closed curve of the region bounded by $y = x^2$ and $y^2 = x$.

Sol The plane $y = x^2$ and $x = y^2$ intersect at $(0,0)$ & $(1,1)$ the positive direction in traversing $(0,1)$.



Along $y = x^2$ we have $dy = 2x dx$ while x from 0 to 1, Thus

$$\int_{x=0}^1 (2x \cdot x^2 - x^4) dx + (x + x^4) 2x dx$$

$$= \int_0^1 (2x^3 - x^4 + 2x^2 + 2x^5) dx = \int_0^1 (2x^5 - x^4 + 2x^3 + 2x^2) dx$$

$$= \left[\frac{x^6}{3} - \frac{x^5}{5} + \frac{x^4}{2} + \frac{2x^3}{3} \right]_0^1 = \frac{1}{3} - \frac{1}{5} + \frac{1}{2} + \frac{2}{3} = \frac{17}{10}$$

Along $u=y^2$, $du = 2y dy$, while y from 1 to 0. ⑤

$$\int_{y=1}^0 (2y^2 \cdot y - y^2) 2y dy + (y^2 + y^2) dy$$

$$= \int_1^0 (4y^4 - 2y^3 + 2y^2) dy = \left. \frac{4y^5}{5} - \frac{2y^4}{4} + \frac{2y^3}{3} \right|_1^0$$

$$= 0 - \left(\frac{4}{5} - \frac{2}{4} + \frac{2}{3} \right) = -\frac{13}{15} - \frac{29}{30}$$

$$\text{Thus } \oint_C (2xy - y^2) dx + (x + y^2) dy = \frac{13}{10} - \frac{29}{30} = \frac{1}{3}$$

By Green's Theorem in the plane.

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$P(x, y) = 2xy - y^2$$

$$Q(x, y) = x + y^2$$

$$\frac{\partial P}{\partial y} = 2x - 2y$$

$$\frac{\partial Q}{\partial x} = 1$$

$$\oint_C (2xy - y^2) dx + (x + y^2) dy = \int_0^1 \int_{x^2}^{\sqrt{x}} (1 - 2x + 2y) dy dx$$

$$= \int_0^1 \left[y - 2xy + y^2 \right]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 \left[(\sqrt{x} - 2x\sqrt{x} + (\sqrt{x})^2) - (x^2 - 2x^3 + x^4) \right] dx$$

$$= \int_0^1 (x^{1/2} - 2x^{3/2} + x - x^2 + 2x^3 - x^4) dx$$

$$= \left[\frac{2}{3} x^{3/2} - \frac{4}{5} x^{5/2} + \frac{x^2}{2} - \frac{x^3}{3} + \frac{2x^4}{4} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{2}{3} - \frac{4}{5} + \frac{1}{2} - \frac{1}{3} + \frac{1}{2} - \frac{1}{5} = \frac{1}{3}$$

Solve using Green's Theorem

$$\oint_C (y + e^{\sqrt{x}}) dx + (2x + \cos(y^2)) dy \quad \text{where } C \text{ is bounded}$$

by $y = x^2$ and $x = y^2$

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dy dx$$

$$P(x, y) = y + e^{\sqrt{x}}$$

$$Q(x, y) = 2x + \cos(y^2)$$

$$\frac{\partial P}{\partial y} = 1$$

$$\frac{\partial Q}{\partial x} = 2$$

$$\oint_C (y + e^{\sqrt{x}}) dx + (2x + \cos(y^2)) dy = \int_0^1 \int_{x^2}^{\sqrt{x}} (2 - 1) dy dx$$

$$= \int_0^1 [y]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 (\sqrt{x} - x^2) dx$$

$$= \left. \frac{2}{3} x^{3/2} - \frac{x^3}{3} \right|_0^1$$

$$= \frac{2}{3} - \frac{1}{3}$$

$$= \frac{1}{3}$$